



Missouri State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Missouri's Learning Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Missouri's Learning Standards for Mathematics to fully support Missouri's students and teachers. The tables on the following pages, organized by Missouri Learning Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Number Sense		
K.NS.A.1	Count to 100 by ones and tens	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.NS.A.2	Count forward beginning from a given number between 1 and 20.	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.NS.A.3	Count backward from a given number between 10 and 1.	Mission 1, Topic H
K.NS.A.4	Read and write numerals and represent a number of objects from 0 to 20.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.NS.B.5	Say the number names when counting objects, in the standard order, pairing each object with one and only one number name and each number name with one and only one object.	Mission 1, Topic A Mission 1, Topic B Mission 5, Topic C
K.NS.B.6	Demonstrate that the last number name said tells the number of objects counted and the number of objects is the same regardless of their arrangement or the order in which they were counted.	Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
K.NS.B.7	Demonstrate that each successive number name refers to a quantity that is one larger than the previous number.	Mission 1, Topic G Mission 1, Topic H

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.NS.B.8	Recognize, without counting, the quantity of groups up to 5 objects arranged in common patterns.	Mission 4, Topic A
K.NS.B.9	Demonstrate that a number can be used to represent "how many" are in a set.	Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 1, Topic H Mission 3, Topic B Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.NS.C.10	Compare two or more sets of objects and identify which set is equal to, more than or less than the other.	Mission 3, Topic B Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H Mission 5, Topic E
K.NS.C.11	Compare two numerals, between 1 and 10, and determine which is more than or less than the other.	Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H
Number Sense & Operations in Base Ten		
K.NBT.A.1	Compose and decompose numbers from 11 to 19 into sets of tens with additional ones.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
Relationships & Algebraic Thinking		
K.RA.A.1	Represent addition and subtraction within 10.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.RA.A.2	Demonstrate fluency for addition and subtraction within 5.	Mission 4, Topic A
K.RA.A.3	Decompose numbers less than or equal to 10 in more than one way.	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G
K.RA.A.4	Make 10 for any number from 1 to 9.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic H
Geometry & Measurement		
K.GM.A.1	Describe several measureable attributes of objects.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
K.GM.A.2	Compare the measurable attributes of two objects.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 3, Topic H
K.GM.B.3	Demonstrate an understanding of concepts of time and devices that measure time.	Grade 1, Mission 5, Topic D
K.GM.B.4	Name the days of the week.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.GM.B.5	Identify pennies, nickels, dimes and quarters.	Grade 1, Mission 6, Topic E
K.GM.C.6	Identify shapes and describe objects in the environment using names of shapes, recognizing the name stays the same regardless of orientation or size.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.GM.C.7	Describe the relative positions of objects in space.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B
K.GM.C.8	Identify and describe the attribute of shapes, and use the attributes to sort a collection of shapes.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.GM.C.9	Draw or model simple two-dimensional shapes.	Mission 6, Topic A
K.GM.C.10	Compose simple shapes to form larger shapes using manipulatives.	Mission 6, Topic B
Data & Statistics		
K.DS.A.1	Classify objects into given categories; count the number of objects in each category.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
K.DS.A.2	Compare category counts using appropriate language.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C

1st Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Number Sense		
1.NS.A.1	Count to 120, starting at any number less than 120.	Mission 4, Topic A Mission 6, Topic B
1.NS.A.2	Read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A Mission 6, Topic B
1.NS.A.3	Count backward from a given number between 20 and 1.	<i>Digital Activities</i> <i>Zearn Math includes digital activities that require counting backwards.</i>
1.NS.A.4	Count by 5s to 100 starting at any multiple of five.	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
Numbers & Operations in Base Ten		
1.NBT.A.1	Understand that 10 can be thought of as a bundle of 10 ones – called a “ten”.	Mission 4, Topic A Mission 6, Topic B
1.NBT.A.2	Understand two-digit numbers are composed of ten(s) and one(s).	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B
1.NBT.A.3	Compare two two-digit numbers using the symbols $>$, $=$ or $<$.	Mission 4, Topic B Mission 6, Topic B
1.NBT.A.4	Count by 10s to 120 starting at any number.	Mission 6, Topic B Mission 6, Topic C
1.NBT.B.5	Add within 100.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 6, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 6, Topic D
1.NBT.B.6	Calculate 10 more or 10 less than a given number mentally without having to count.	Mission 2, Topic D Mission 4, Topic A Mission 6, Topic B
1.NBT.B.7	Add or subtract a multiple of 10 from another twodigit number, and justify the solution.	Mission 4, Topic C Mission 6, Topic C
Relationships & Algebraic Thinking		
1.RA.A.1	Use addition and subtraction within 20 to solve problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
1.RA.A.2	Solve problems that call for addition of three whole numbers whose sum is within 20.	Mission 2, Topic A
1.RA.A.3	Develop the meaning of the equal sign and determine if equations involving addition and subtraction are true or false.	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic C
1.RA.A.4	Determine the unknown whole number in an addition or subtraction equation relating three whole numbers.	Mission 1, Topic D Mission 1, Topic H Mission 2, Topic C
1.RA.B.5	Use properties as strategies to add and subtract.	Mission 1, Topic E Mission 1, Topic F Mission 2, Topic A Mission 2, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.RA.B.6	Demonstrate that subtraction can be solved as an unknown-addend problem.	Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.RA.C.7	Add and subtract within 20.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D
1.RA.C.8	Demonstrate fluency with addition and subtraction within 10.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D
Geometry & Measurement		
1.GM.A.1	Distinguish between defining attributes versus non-defining attributes; build and draw shapes that possess defining attributes.	Mission 5, Topic A
1.GM.A.2	Compose and decompose two- and three-dimensional shapes to build an understanding of part-whole relationships and the properties of the original and composite shapes.	Mission 5, Topic B
1.GM.A.3	Recognize two- and three-dimensional shapes from different perspectives and orientations.	Kindergarten, Mission 2, Topic A Kindergarten, Mission 2, Topic B Kindergarten, Mission 2, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.GM.A.4	Partition circles and rectangles into two or four equal shares, and describe the shares and the wholes verbally.	Mission 5, Topic C Mission 5, Topic D
1.GM.B.5	Order three or more objects by length.	Mission 3, Topic A Mission 3, Topic B
1.GM.B.6	Compare the lengths of two objects indirectly by using a third object.	Mission 3, Topic A Mission 3, Topic B
1.GM.B.7	Demonstrate the ability to measure length or distance using objects.	Mission 3, Topic B Mission 3, Topic C
1.GM.C.8	Tell and write time in hours and half-hours using analog and digital clocks.	Mission 5, Topic D
1.GM.C.9	Know the value of a penny, nickel, dime and quarter.	Mission 6, Topic E
Data & Statistics		
1.DS.A.1	Collect, organize and represent data with up to three categories.	Mission 3, Topic D
1.DS.A.1	Draw conclusions from object graphs, picture graphs, T-charts and tallies.	Mission 3, Topic D

2nd Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Numbers Sense & Operations in Base Ten		
2.NBT.A.1	Understand three-digit numbers are composed of hundreds, tens and ones.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.A.2	Understand that 100 can be thought of as 10 tens – called a “hundred”.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.A.3	Count within 1000 by 1s, 10s and 100s starting with any number.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.NBT.A.4	Read and write numbers to 1000 using number names, base-ten numerals and expanded form.	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.A.5	Compare two three-digit numbers using the symbols $>$, $=$ or $<$.	Mission 3, Topic E Mission 3, Topic F
2.NBT.B.6	Demonstrate fluency with addition and subtraction within 100.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.NBT.B.7	Add up to four two-digit numbers.	Mission 4, Topic D Mission 6, Topic A Mission 7, Topic B Mission 8, Topic D
2.NBT.B.8	Add or subtract within 1000, and justify the solution.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
2.NBT.B.9	Use the relationship between addition and subtraction to solve problems.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
2.NBT.B.10	Add or subtract mentally 10 or 100 to or from a given number within 1000.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.NBT.C.11	Write and solve problems involving addition and subtraction within 100.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
Relationships & Algebraic Thinking		
2.RA.A.1	Demonstrate fluency with addition and subtraction within 20.	Mission 1, Topic A Mission 1, Topic B
2.RA.B.2	Determine if a set of objects has an odd or even number of members. a. Count by 2s to 100 starting	Mission 6, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	with any even number. b. Express even numbers as pairings/groups of 2, and write an expression to represent the number using addends of 2. c. Express even numbers as being composed of equal groups and write an expression to represent the number with 2 equal addends.	
2.RA.B.3	Find the total number of objects arranged in a rectangular array with up to 5 rows and 5 columns, and write an equation to represent the total as a sum of equal addends.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
Geometry & Measurement		
2.GM.A.1	Recognize and draw shapes having specified attributes, such as a given number of angles or sides. a. Identify triangles, quadrilaterals, pentagons, hexagons, circles and cubes. b. Identify the faces of three-dimensional objects.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C
2.GM.A.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of squares.	Mission 6, Topic C
2.GM.A.3	Partition circles and rectangles into two, three or four equal shares, and describe the shares and the whole. a. Demonstrate that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
2.GM.B.4	Measure the length of an object by selecting and using appropriate tools.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D Mission 7, Topic F Mission 8, Topic A
2.GM.B.5	Analyze the results of measuring the same object with different units.	Mission 2, Topic C Mission 7, Topic D
2.GM.B.6	Estimate lengths using units of inches, feet, yards, centimeters and meters.	Mission 2, Topic B Mission 2, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 7, Topic D
2.GM.B.7	Measure to determine how much longer one object is than another.	Mission 2, Topic C Mission 2, Topic D Mission 7, Topic D
2.GM.C.8	Use addition and subtraction within 100 to solve problems involving lengths that are given in the same units.	Mission 2, Topic D Mission 7, Topic E Mission 7, Topic F
2.GM.C.9	Represent whole numbers as lengths on a number line, and represent whole-number sums and differences within 100 on a number line.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.GM.D.10	Tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m.	Mission 8, Topic D
2.GM.D.11	Describe a time shown on a digital clock as representing hours and minutes, and relate a time shown on a digital clock to the same time on an analog clock.	Mission 8, Topic D
2.GM.D.12	Find the value of combinations of dollar bills, quarters, dimes, nickels and pennies, using \$ and ¢ appropriately.	Mission 3, Topic D Mission 7, Topic B
2.GM.D.13	Find combinations of coins that equal a given amount.	Mission 7, Topic B
Data & Statistics		
2.DS.A.1	Create a line plot to represent a set of numeric data, given a horizontal scale marked in whole numbers.	Mission 7, Topic F
2.DS.A.2	Generate measurement data to the nearest whole unit, and display the data in a line plot.	Mission 7, Topic F
2.DS.A.3	Draw a picture graph or a bar graph to represent a data set with up to four categories.	Mission 7, Topic A
2.DS.A.4	Solve problems using information presented in line plots, picture graphs and bar graphs.	Mission 7, Topic A

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.DS.A.5	Draw conclusions from line plots, picture graphs and bar graphs.	Mission 7, Topic A Mission 7, Topic F

3rd Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Numbers Sense & Operations in Base Ten		
3.NBT.A.1	Round whole numbers to the nearest 10 or 100.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.NBT.A.2	Read, write and identify whole numbers within 100,000 using base ten numerals, number names and expanded form.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.NBT.A.3	Demonstrate fluency with addition and subtraction within 1000.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.NBT.A.4	Multiply whole numbers by multiples of 10 in the range 10-90.	Mission 3, Topic F
Numbers Sense & Operations in Fractions		
3.NF.A.1	Understand a unit fraction as the quantity formed by one part when a whole is partitioned into equal parts.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.NF.A.2	Understand that when a whole is partitioned equally, a fraction can be used to represent a portion of the whole. a. Describe the numerator as representing the number of pieces being considered. b. Describe the denominator as the number of pieces that make the whole.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.NF.A.3	Represent fractions on a number line. a. Understand the whole is the interval from 0 to 1. b. Understand the whole is partitioned into equal parts. c. Understand a fraction represents the endpoint of the length a given number of partitions from 0.	Mission 5, Topic D
3.NF.A.4	Demonstrate that two fractions are equivalent if they are the same size, or the same point on a number line.	Mission 5, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.NF.A.5	Recognize and generate equivalent fractions using visual models, and justify why the fractions are equivalent.	Mission 5, Topic E
3.NF.A.6	Compare two fractions with the same numerator or denominator using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic F
3.NF.A.7	Explain why fraction comparisons are only valid when the two fractions refer to the same whole.	Mission 5, Topic C Mission 5, Topic F
Relationships & Algebraic Thinking		
3.RA.A.1	Interpret products of whole numbers.	Mission 1, Topic A Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.RA.A.2	Interpret quotients of whole numbers.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.RA.A.3	Describe in words or drawings a problem that illustrates a multiplication or division situation.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 3, Topic E
3.RA.A.4	Use multiplication and division within 100 to solve problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.RA.A.5	Determine the unknown number in a multiplication or division equation relating three whole numbers.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.RA.B.6	Apply properties of operations as strategies to multiply and divide.	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
3.RA.C.7	Multiply and divide with numbers and results within 100 using strategies such as the relationship between multiplication and division or properties of operations. Know all products of two one-digit numbers.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.RA.C.8	Demonstrate fluency with products within 100.	Mission 1, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.RA.D.9	Write and solve two-step problems involving variables using any of the four operations.	Mission 1, Topic F Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F Mission 7, Topic A
3.RA.D.10	Interpret the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 1, Topic F Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F
3.RA.E.11	Identify arithmetic patterns and explain the patterns using properties of operations.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
Geometry & Measurement		
3.GM.A.1	Understand that shapes in different categories may share attributes and that the shared attributes can define a larger category.	Mission 7, Topic B
3.GM.A.2	Distinguish rhombuses and rectangles as examples of quadrilaterals, and draw examples of quadrilaterals that do not belong to these subcategories.	Mission 7, Topic B
3.GM.A.3	Partition shapes into parts with equal areas, and express the area of each part as a unit fraction of the whole.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.GM.B.4	Tell and write time to the nearest minute.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.GM.B.5	Estimate time intervals in minutes.	Mission 2, Topic C Mission 2, Topic D
3.GM.B.6	Solve problems involving addition and subtraction of minutes.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.GM.B.7	Measure or estimate length, liquid volume and weight of objects.	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.GM.B.8	Use the four operations to solve problems involving lengths, liquid volumes or weights given in the same units.	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.GM.C.9	Calculate area by using unit squares to cover a plane figure with no gaps or overlaps.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.GM.C.10	Label area measurements with squared units.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D
3.GM.C.11	Demonstrate that tiling a rectangle to find the area and multiplying the side lengths result in the same value.	Mission 4, Topic A Mission 4, Topic B
3.GM.C.12	Multiply whole-number side lengths to solve problems involving the area of rectangles.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.GM.C.13	Find rectangular arrangements that can be formed for a given area.	Mission 4, Topic C Mission 4, Topic D
3.GM.C.14	Decompose a rectangle into smaller rectangles to find the area of the original rectangle.	Mission 4, Topic D
3.GM.D.15	Solve problems involving perimeters of polygons.	Mission 7, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 7, Topic D Mission 7, Topic E
3.GM.D.16	Understand that rectangles can have equal perimeters but different areas, or rectangles can have equal areas but different perimeters.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
Data & Statistics		
3.DS.A.1	Create frequency tables, scaled picture graphs and bar graphs to represent a data set with several categories	Mission 6, Topic A
3.DS.A.2	Solve one- and two-step problems using information presented in bar and/or picture graphs.	Mission 6, Topic A
3.DS.A.3	Create a line plot to represent data.	Mission 6, Topic B Mission 7, Topic D
3.DS.A.4	Use data shown in a line plot to answer questions.	Mission 6, Topic B Mission 7, Topic D

4th Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Numbers Sense & Operations in Base Ten		
4.NBT.A.1	Round multi-digit whole numbers to any place.	Mission 1, Topic C
4.NBT.A.2	Read, write and identify multi-digit whole numbers up to one million using number names, base ten numerals and expanded form.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.A.3	Compare two multi-digit numbers using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 1, Topic B
4.NBT.A.4	Understand that in a multi-digit whole number, a digit represents 10 times what it would represent in the place to its right.	Mission 1, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic G
4.NBT.A.5	Demonstrate fluency with addition and subtraction of whole numbers.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.A.6	Multiply a whole number of up to four digits by a one-digit whole number and multiply two twodigit numbers, and justify the solution.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.NBT.A.7	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, and justify the solution.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
Numbers Sense & Operations in Fractions		
4.NF.A.1	Explain and/or illustrate why two fractions are equivalent.	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.NF.A.2	Recognize and generate equivalent fractions.	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B
4.NF.A.3	Compare two fractions using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 5, Topic C Mission 5, Topic E
4.NF.B.4	Understand addition and subtraction of fractions as joining/composing and separating/decomposing parts referring to the same whole.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.NF.B.5	Decompose a fraction into a sum of fractions with the same denominator and record each decomposition with an equation and justification.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
4.NF.B.6	Solve problems involving adding and subtracting fractions and mixed numbers with like denominators.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
4.NF.B.7	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number.	Mission 5, Topic A Mission 5, Topic E Mission 5, Topic G
4.NF.B.8	Solve problems involving multiplication of a fraction by a whole number.	Mission 5, Topic A Mission 5, Topic E Mission 5, Topic G
4.NF.C.9	Use decimal notation for fractions with denominators of 10 or 100.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.NF.C.10	Understand that fractions and decimals are equivalent representations of the same quantity.	Mission 6, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.NF.C.11	Read, write and identify decimals to the hundredths place using number names, base ten numerals and expanded form.	Mission 6, Topic B
4.NF.C.12	Compare two decimals to the hundredths place using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 6, Topic B Mission 6, Topic C
Relationships & Algebraic Thinking		
4.RA.A.1	Multiply or divide to solve problems involving a multiplicative comparison.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B
4.RA.A.2	Solve multi-step whole number problems involving the four operations and variables and using estimation to interpret the reasonableness of the answer.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 5, Topic H Mission 7, Topic B Mission 7, Topic C
4.RA.A.3	Solve whole number division problems involving variables in which remainders need to be interpreted, and justify the solution.	Mission 3, Topic E Mission 3, Topic G
4.RA.B.4	Recognize that a whole number is a multiple of each of its factors and find the multiples for a given whole number.	Mission 3, Topic F
4.RA.B.5	Determine if a whole number within 100 is composite or prime, and find all factor pairs for whole numbers within 100.	Mission 3, Topic F
4.RA.C.6	Generate a number pattern that follows a given rule.	Mission 3, Topic H

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.RA.C.7	Use words or mathematical symbols to express a rule for a given pattern.	Mission 3, Topic H
Geometry & Measurement		
4.GM.A.1	Draw and identify points, lines, line segments, rays, angles, perpendicular lines and parallel lines.	Mission 4, Topic A Mission 4, Topic D
4.GM.A.2	Classify two-dimensional shapes by their sides and/or angles.	Mission 4, Topic D
4.GM.A.3	Construct lines of symmetry for a twodimensional figure.	Mission 4, Topic D
4.GM.B.4	Identify and estimate angles and their measure.	Mission 4, Topic B
4.GM.B.5	Draw and measure angles in whole-number degrees using a protractor.	Mission 4, Topic B
4.GM.C.6	Know relative sizes of measurement units within one system of units. a. Convert measurements in a larger unit in terms of a smaller unit.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.GM.C.7	Use the four operations to solve problems involving distances, intervals of time, liquid volume, weight of objects and money.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic C Mission 6, Topic E Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.GM.C.8	Apply the area and perimeter formulas for rectangles to solve problems.	Mission 3, Topic A Mission 3, Topic H
Data & Statistics		
4.DS.A.1	Create a frequency table and/or line plot to display measurement data.	Mission 5, Topic E Mission 5, Topic G
4.DS.A.2	Solve problems involving addition and subtraction by using information presented in a data display.	Mission 5, Topic E Mission 5, Topic G
4.DS.A.3	Analyze the data in a frequency table, line plot, bar graph or picture graph.	Grade 3, Mission 6, Topic A Mission 5, Topic E Mission 5, Topic G

5th Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Numbers Sense & Operations in Base Ten		
5.NBT.A.1	Read, write and identify numbers from billions to thousandths using number names, base ten numerals and expanded form.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
5.NBT.A.2	Compare two numbers from billions to thousandths using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 1, Topic B
5.NBT.A.3	Understand that in a multi-digit number, a digit represents $\frac{1}{10}$ times what it would represent in the place to its left.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
5.NBT.A.4	Evaluate the value of powers of 10 and understand the relationship to the place value system.	Mission 1, Topic A Mission 1, Topic D Mission 1, Topic E Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
5.NBT.A.5	Round numbers from billions to thousandths place.	Mission 1, Topic C
5.NBT.A.6	Add and subtract multidigit whole numbers and decimals to the thousandths place, and justify the solution.	Mission 1, Topic D
5.NBT.A.7	Multiply multi-digit whole numbers and decimals to the hundredths place, and justify the solution.	Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic D Mission 4, Topic E Mission 4, Topic F

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.NBT.A.8	Divide multi-digit whole numbers and decimals to the hundredths place using up to two-digit divisors and four-digit dividends, and justify the solution.	Mission 1, Topic F Mission 2, Topic C Mission 2, Topic E Mission 2, Topic F Mission 2, Topic G Mission 4, Topic G
Numbers Sense & Operations in Fractions		
5.NF.A.1	Understand that parts of a whole can be expressed as fractions and/or decimals.	Mission 1, Topic B
5.NF.A.2	Convert decimals to fractions and fractions to decimals.	Mission 1, Topic B Mission 4, Topic E
5.NF.A.3	Compare and order fractions and/or decimals to the thousandths place using the symbols $>$, $=$ or $<$, and justify the solution.	Mission 1, Topic B
5.NF.B.4	Estimate results of sums, differences and products with fractions and decimals to the thousandths.	Mission 2, Topic C Mission 3, Topic D Mission 3, Topic D
5.NF.B.5	Justify the reasonableness of a product when multiplying with fractions. a. Estimate the size of the product based on the size of the two factors. b. Explain why multiplying a given number by a fraction greater than 1 results in a product larger than the given number. c. Explain why multiplying a given number by a fraction less than 1 results in a product smaller than the given number. d. Explain why multiplying the numerator and denominator by the same number is equivalent to multiplying the fraction by 1.	Mission 4, Topic F
5.NF.B.6	Solve problems involving addition and subtraction of fractions and mixed numbers with unlike denominators, and justify the solution.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D
5.NF.B.7	Extend the concept of multiplication to multiply a fraction or whole number by a fraction. a. Recognize the relationship between multiplying fractions and finding the areas of rectangles with fractional side	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	lengths. b. Calculate and interpret the product of a fraction by a whole number and a whole number by a fraction. c. Calculate and interpret the product of two fractions less than one.	
5.NF.B.8	Extend the concept of division to divide unit fractions and whole numbers by using visual fraction models and equations. a. Calculate and interpret the quotient of a unit fraction by a non-zero whole number. b. Calculate and interpret the quotient of a whole number by a unit fraction.	Mission 4, Topic G
Relationships & Algebraic Thinking		
5.RA.A.1	Investigate the relationship between two numeric patterns. a. Generate two numeric patterns given two rules. b. Translate two numeric patterns into two sets of ordered pairs. c. Graph numeric patterns on the Cartesian coordinate plane. d. Identify the relationship between two numeric patterns.	Mission 6, Topic B Mission 6, Topic D
5.RA.A.2	Write a rule to describe or explain a given numeric pattern.	Mission 6, Topic B
5.RA.B.3	Write, evaluate and interpret numeric expressions using the order of operations.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
5.RA.B.4	Translate written expressions into algebraic expressions.	Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic H Mission 6, Topic B
5.RA.B.5	Solve and justify multistep problems involving variables, whole numbers, fractions and decimals.	Mission 3, Topic B Mission 3, Topic D Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
		Mission 4, Topic F Mission 4, Topic H
Geometry & Measurement		
5.GM.A.1	Understand that attributes belonging to a category of figures also belong to all subcategories.	Mission 5, Topic D
5.GM.A.2	Classify figures in a hierarchy based on properties.	Mission 5, Topic D
5.GM.A.3	Analyze and describe the properties of prisms and pyramids.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.GM.B.4	Understand the concept of volume and recognize that volume is measured in cubic units. a. Describe a cube with edge length 1 unit as a “unit cube” and is said to have “one cubic unit” of volume and can be used to measure volume. b. Understand that the volume of a right rectangular prism can be found by stacking multiple layers of the base.	Mission 5, Topic A Mission 5, Topic B
5.GM.B.5	Apply the formulas $V = l \times w \times h$ and $V = B \times h$ for volume of right rectangular prisms with whole-number edge lengths.	Mission 5, Topic B
5.GM.C.6	Define a first quadrant Cartesian coordinate system. a. Represent the axes as scaled perpendicular number lines that both intersect at 0, the origin. b. Identify any point on the Cartesian coordinate plane by its ordered pair coordinates. c. Define the first number in an ordered pair as the horizontal distance from the origin. d. Define the second number in an ordered pair as the vertical distance from the origin.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
5.GM.C.7	Plot and interpret points in the first quadrant of the Cartesian coordinate plane.	Mission 6, Topic C Mission 6, Topic D
5.GM.D.8	Convert measurements of capacity, length and weight within a given measurement system.	Mission 1, Topic A Mission 2, Topic D Mission 4, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.GM.D.9	Solve multi-step problems that require measurement conversions.	Mission 4, Topic C Mission 4, Topic E
Data & Statistics		
5.DS.A.1	Create a line graph to represent a data set, and analyze the data to answer questions and solve problems.	Mission 6, Topic D
5.DS.A.2	Create a line plot to represent a given or generated data set, and analyze the data to answer questions and solve problems, recognizing the outliers and generating the median.	Mission 4, Topic A

6th Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.A.1	Understand a ratio as a comparison of two quantities and represent these comparisons.	Mission 2, Topic A Mission 2, Topic B
6.RP.A.2	Understand the concept of a unit rate associated with a ratio, and describe the meaning of unit rate.	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.RP.A.3	Solve problems involving ratios and rates. a. Create tables of equivalent ratios, find missing values in the tables and plot the pairs of values on the Cartesian coordinate plane. b. Solve unit rate problems. c. Solve percent problems. d. Convert measurement units within and between two systems of measurement.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 2, Topic F Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 6, Topic B Mission 6, Topic D
Number Sense & Operations		
6.NS.A.1	Compute and interpret quotients of positive fractions. a. Solve problems involving division of fractions by fractions.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E
6.NS.B.2	Demonstrate fluency with division of multi-digit whole numbers.	Mission 5, Topic D
6.NS.B.3	Demonstrate fluency with addition, subtraction, multiplication and division of decimals.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 6, Topic A Mission 8, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.NS.B.4	Find common factors and multiples. a. Find the greatest common factor (GCF) and the least common multiple (LCM). b. Use the distributive property to express a sum of two whole numbers with a common factor as a multiple of a sum of two whole numbers.	Mission 7, Topic D
6.NS.C.5	Use positive and negative numbers to represent quantities.	Mission 7, Topic A
6.NS.C.6	Locate a rational number as a point on the number line. a. Locate rational numbers on a horizontal or vertical number line. b. Write, interpret and explain problems of ordering of rational numbers. c. Understand that a number and its opposite (additive inverse) are located on opposite sides of zero on the number line.	Mission 7, Topic A Mission 7, Topic C
6.NS.C.7	Understand that the absolute value of a rational number is its distance from 0 on the number line.	Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
6.NS.C.8	Extend prior knowledge to generate equivalent representations of rational numbers between fractions, decimals and percentages (limited to terminating decimals and/or benchmark fractions of $\frac{1}{3}$ and $\frac{2}{3}$).	Mission 3, Topic D
Expressions, Equations & Inequalities		
6.EE.A.1	Describe the difference between an expression and an equation.	Mission 6, Topic B
6.EE.A.2	Create and evaluate expressions involving variables and whole number exponents. a. Identify parts of an expression using mathematical terminology. b. Evaluate expressions at specific values of the variables. c. Evaluate non-negative rational number expressions. d. Write and evaluate algebraic expressions. e. Understand the meaning of the variable in the context of the situation.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic F Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE.A.3	Identify and generate equivalent algebraic expressions using mathematical properties.	Mission 5, Topic D Mission 6, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.EE1.B.4	Use substitution to determine whether a given number in a specified set makes a one-variable equation or inequality true.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE1.B.5	Understand that if any solutions exist, the solution set for an equation or inequality consists of values that make the equation or inequality true.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE1.B.6	Write and solve equations using variables to represent quantities, and understand the meaning of the variable in the context of the situation.	Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B
6.EE1.B.7	Solve one-step linear equations in one variable involving non-negative rational numbers.	Mission 6, Topic A Mission 6, Topic B
6.EE1.B.8	Recognize that inequalities may have infinitely many solutions. a. Write an inequality of the form $x > c$, $x < c$, $x \geq c$, or $x \leq c$ to represent a constraint or condition. b. Graph the solution set of an inequality.	Mission 7, Topic B
6.EE1.C.9	Identify and describe relationships between two variables that change in relationship to one another. a. Write an equation to express one quantity, the dependent variable, in terms of the other quantity, the independent variable. b. Analyze the relationship between the dependent and independent variables using graphs, tables and equations and relate these representations to each other.	Mission 6, Topic D
Geometry & Measurement		
6.GM.A.1	Find the area of polygons by composing or decomposing the shapes into rectangles or triangles.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 4, Topic D
6.GM.A.2	Find the volume of right rectangular prisms. a. Understand that the volume of a right rectangular prism can be found by filling the prism with multiple	Mission 1, Topic E Mission 4, Topic D Mission 4, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	layers of the base. b. Apply $V = l * w * h$ and $V = Bh$ to find the volume of right rectangular prisms.	
6.GM.A.3	Solve problems by graphing points in all four quadrants of the Cartesian coordinate plane. a. Understand signs of numbers in ordered pairs as indicating locations in quadrants of the Cartesian coordinate plane b. Recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. c. Find distances between points with the same first coordinate or the same second coordinate. d. Construct polygons in the Cartesian coordinate plane.	Mission 7, Topic C
6.GM.A.4	Solve problems using nets. a. Represent three-dimensional figures using nets made up of rectangles and triangles. b. Use nets to find the surface area of threedimensional figures whose sides are made up of rectangles and triangles.	Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G
Data Analysis, Statistics & Probability		
6.DSP.A.1	Recognize a statistical question as one that anticipates variability in the data related to the question and accounts for it in the answers.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic D
6.DSP.A.2	Understand that a set of data collected to answer a statistical question has a distribution which can be described by its center, spread and overall shape.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic E
6.DSP.A.3	Recognize that a measure of center for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary from a single number.	Mission 8, Topic B
6.DSP.B.1	Display and interpret data. a. Use dot plots, histograms and box plots to display and interpret numerical data. b. Create and interpret circle graphs.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
6.DSP.B.2	Summarize numerical data sets in relation to the context. a. Report the number of observations. b. Describe the nature of the attribute under	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS	ZEARN MATH
6th Grade Standards	Lessons
investigation, including how it was measured and its units of measurement. c. Give quantitative measures of center (median and/or mean) and variability (interquartile range and/or mean absolute deviation), as well as describing any overall pattern and any striking deviations from the overall pattern with reference to the context of the data. d. Analyze the choice of measures of center and variability based on the shape of the data distribution and/or the context of the data.	Mission 8, Topic D Mission 8, Topic E

7th Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.A.1	Compute unit rates, including those that involve complex fractions, with like or different units.	Mission 2, Topic C Mission 4, Topic A
7.RP.A.2	Recognize and represent proportional relationships between quantities. a. Determine when two quantities are in a proportional relationship. b. Identify and/or compute the constant of proportionality (unit rate). c. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation. d. Recognize that the graph of any proportional relationship will pass through the origin.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 3, Topic A Mission 3, Topic B Mission 4, Topic A Mission 5, Topic C Mission 5, Topic D
7.RP.A.3	Solve problems involving ratios, rates, percentages and proportional relationships.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
Number Sense & Operations		
7.NS.A.1	Apply and extend previous understandings of numbers to add and subtract rational numbers. a. Add and subtract rational numbers. b. Represent addition and subtraction on a horizontal or vertical number line. c. Describe situations and show that a number and its opposite have a sum of 0 (additive inverses). d. Understand subtraction of rational numbers as adding the additive inverse. e. Determine the distance between two rational numbers on the number line is the absolute value of their difference. f. Interpret sums and differences of rational numbers.	Mission 5, Topic A Mission 5, Topic B Mission 6, Topic D Mission 7, Topic B
7.NS.A.2	Apply and extend previous understandings of numbers to multiply and divide rational numbers. a. Multiply and divide rational numbers. b. Determine that a number and its reciprocal have a product of 1	Mission 4, Topic A Mission 5, Topic A Mission 5, Topic C Mission 8, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	(multiplicative inverse). c. Understand that every quotient of integers (with non-zero divisor) is a rational number. d. Convert a rational number to a decimal. e. Understand that all rational numbers can be written as fractions or decimal numbers that terminate or repeat. f. Interpret products and quotients of rational numbers by describing real-world contexts.	
7.NS.A.3	Solve problems involving the four arithmetic operations with rational numbers.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Expressions, Equations & Inequalities		
7.EE1.A.1	Apply properties of operations to simplify and to factor linear algebraic expressions with rational coefficients.	Mission 6, Topic D
7.EE1.A.2	Understand how to use equivalent expressions to clarify quantities in a problem.	Mission 6, Topic B
7.EE1.B.3	Solve multi-step problems posed with rational numbers. a. Convert between equivalent forms of the same number. b. Assess the reasonableness of answers using mental computation and estimation strategies.	Mission 3, Topic C Mission 5, Topic C Mission 5, Topic F Mission 6, Topic A Mission 6, Topic B
7.EE1.B.4	Write and/or solve linear equations and inequalities in one variable. a. Write and/or solve equations of the form $x+p=q$ and $px=q$ in which p and q are rational numbers. b. Write and/or solve two-step equations of the form $px+q=r$ and $p(x+q)=r$, where p , q and r are rational numbers, and interpret the meaning of the solution in the context of the problem. c. Write, solve and/or graph inequalities of the form $px+q>r$ or $px+q<r$, where p , q and r are rational numbers.	Mission 5, Topic E Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic A
Geometry & Measurement		

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.GM.A.1	Solve problems involving scale drawings of real objects and geometric figures, including computing actual lengths and areas from a scale drawing and reproducing the drawing at a different scale.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 2, Topic A Mission 3, Topic B Mission 3, Topic C
7.GM.A.2	Use a variety of tools to construct geometric shapes. a. Determine if provided constraints will create a unique triangle through construction. b. Construct special quadrilaterals given specific parameters.	Mission 3, Topic A Mission 7, Topic B Mission 7, Topic C
7.GM.A.3	Describe two-dimensional cross sections of pyramids, prisms, cones and cylinders.	Mission 7, Topic C
7.GM.A.4	Understand concepts of circles. a. Analyze the relationships among the circumference, the radius, the diameter, the area and Pi in a circle. b. Know and apply the formulas for circumference and area of circles to solve problems.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.GM.B.5	Use angle properties to write and solve equations for an unknown angle.	Mission 7, Topic A
7.GM.B.6	Understand the relationship between area, surface area and volume. a. Find the area of triangles, quadrilaterals and other polygons composed of triangles and rectangles. b. Find the volume and surface area of prisms, pyramids and cylinders.	Mission 1, Topic A Mission 2, Topic C Mission 3, Topic B Mission 7, Topic C Mission 7, Topic D
Data Analysis, Statistics & Probability		
7.DSP.A.1	Understand that statistics can be used to gain information about a population by examining a sample of the population. a. Understand that a sample is a subset of a population. b. Understand that generalizations from a sample are valid only if the sample is representative of the population. c. Understand that random sampling is used to produce representative samples and support valid inferences.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.DSP.A.2	Use data from multiple samples to draw inferences about a population and investigate variability in estimates of the characteristic of interest.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.DSP.B.3	Analyze different data distributions using statistical measures.	Mission 8, Topic C Mission 8, Topic D
7.DSP.B.4	Compare the numerical measures of center, measures of frequency and measures of variability from two random samples to draw inferences about the population.	Mission 8, Topic D Mission 8, Topic E
7.DSP.C.5	Investigate the probability of chance events. a. Determine probabilities of simple events. b. Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring.	Mission 8, Topic A
7.DSP.C.6	Investigate the relationship between theoretical and experimental probabilities for simple events. a. Predict outcomes using theoretical probability. b. Perform experiments that model theoretical probability. c. Compare theoretical and experimental probabilities.	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E
7.DSP.C.7	Explain possible discrepancies between a developed probability model and observed frequencies. a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. b. Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E
7.DSP.C.8	Find probabilities of compound events using organized lists, tables, tree diagrams and simulations. a. Represent the sample space of a compound event. b. Design and use a simulation to generate frequencies for compound events.	Mission 8, Topic A Mission 8, Topic B

8th Grade

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
Number Sense & Operations		
8.NS.A.1	Explore the real number system. a. Know the differences between rational and irrational numbers. b. Understand that all rational numbers have a decimal expansion that terminates or repeats. c. Convert decimals which repeat into fractions and fractions into repeating decimals. d. Generate equivalent representations of rational numbers.	Mission 8, Topic E
8.NS.A.2	Estimate the value and compare the size of irrational numbers and approximate their locations on a number line.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
Expressions, Equations & Inequalities		
8.EEI.A.1	Know and apply the properties of integer exponents to generate equivalent expressions.	Mission 7, Topic B Mission 7, Topic C
8.EEI.A.2	Investigate concepts of square and cube roots. a. Solve equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. b. Evaluate square roots of perfect squares less than or equal to 625 and cube roots of perfect cubes less than or equal to 1000. c. Recognize that square roots of non-perfect squares are irrational.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
8.EEI.A.3	Express very large and very small quantities in scientific notation and approximate how many times larger one is than the other.	Mission 7, Topic C Mission 7, Topic D
8.EEI.A.4	Use scientific notation to solve problems. a. Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. b. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities.	Mission 7, Topic C Mission 7, Topic D

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.EE1.B.5	Graph proportional relationships. a. Interpret the unit rate as the slope of the graph. b. Compare two different proportional relationships.	Mission 3, Topic A Mission 3, Topic B
8.EE1.B.6	Apply concepts of slope and y-intercept to graphs, equations and proportional relationships. a. Explain why the slope (m) is the same between any two distinct points on a non-vertical line in the Cartesian coordinate plane. b. Derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b.	Mission 2, Topic C Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
8.EE1.C.7	Solve linear equations and inequalities in one variable. a. Create and identify linear equations with one solution, infinitely many solutions or no solutions. b. Solve linear equations and inequalities with rational number coefficients, including equations and inequalities whose solutions require expanding expressions using the distributive property and combining like terms.	Mission 4, Topic B
8.EE1.C.8	Analyze and solve systems of linear equations. a. Graph systems of linear equations and recognize the intersection as the solution to the system. b. Explain why solution(s) to a system of two linear equations in two variables correspond to point(s) of intersection of the graphs. c. Explain why systems of linear equations can have one solution, no solution or infinitely many solutions. d. Solve systems of two linear equations.	Mission 3, Topic D Mission 3, Topic E Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
Geometry & Measurements		
8.GM.A.1	Verify experimentally the congruence properties of rigid transformations. a. Verify that angle measure, betweenness, collinearity and distance are preserved under rigid transformations. b. Investigate if orientation is preserved under rigid transformations.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 3, Topic B
8.GM.A.2	Understand that two-dimensional figures are congruent if a series of rigid transformations can be performed to map the pre-image to the image. a. Describe a possible sequence of rigid transformations between two congruent figures.	Mission 1, Topic C Mission 1, Topic D Mission 2, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.GM.A.3	Describe the effect of dilations, translations, rotations and reflections on two-dimensional figures using coordinates.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C
8.GM.A.4	Understand that two-dimensional figures are similar if a series of transformations (rotations, reflections, translations and dilations) can be performed to map the pre-image to the image. a. Describe a possible sequence of transformations between two similar figures.	Mission 2, Topic B
8.GM.A.5	Explore angle relationships and establish informal arguments. a. Derive the sum of the interior angles of a triangle. b. Explore the relationship between the interior and exterior angles of a triangle. c. Construct and explore the angles created when parallel lines are cut by a transversal. d. Use the properties of similar figures to solve problems.	Mission 1, Topic D Mission 2, Topic B Mission 2, Topic D
8.GM.B.6	Use models to demonstrate a proof of the Pythagorean Theorem and its converse.	Mission 8, Topic C
8.GM.B.7	Use the Pythagorean Theorem to determine unknown side lengths in right triangles in problems in two- and three-dimensional contexts.	Mission 8, Topic C
8.GM.B.8	Use the Pythagorean Theorem to find the distance between points in a Cartesian coordinate system.	Mission 8, Topic C
8.GM.C.9	Solve problems involving surface area and volume. a. Understand the concept of surface area and find surface area of pyramids. b. Understand the concepts of volume and find the volume of pyramids, cones and spheres.	Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Data Analysis, Statistics & Probability		
8.DSP.A.1	Construct and interpret scatter plots of bivariate measurement data to investigate patterns of association between two quantities.	Mission 6, Topic A Mission 6, Topic B
8.DSP.A.2	Generate and use a trend line for bivariate data, and informally assess the fit of the line.	Mission 6, Topic B

MISSOURI'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.DSP.A.3	Interpret the parameters of a linear model of bivariate measurement data to solve problems.	Mission 6, Topic B
8.DSP.A.4	Understand the patterns of association in bivariate categorical data displayed in a two-way table. a. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. b. Use relative frequencies calculated for rows or columns to describe possible association between the two variables.	Mission 6, Topic C
Functions		
8.F.A.1	Explore the concept of functions. (The use of function notation is not required.) a. Understand that a function assigns to each input exactly one output. b. Determine if a relation is a function. c. Graph a function.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E
8.F.A.2	Compare characteristics of two functions each represented in a different way.	Mission 5, Topic B Mission 5, Topic C
8.F.A.3	Investigate the differences between linear and nonlinear functions. a. Interpret the equation $y = mx + b$ as defining a linear function, whose parameters are the slope (m) and the y-intercept (b). b. Recognize that the graph of a linear function has a constant rate of change c. Give examples of nonlinear functions.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
8.F.B.1	Use functions to model linear relationships between quantities. a. Explain the parameters of a linear function based on the context of a problem. b. Determine the parameters of a linear function. c. Determine the x-intercept of a linear function.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
8.F.B.2	Describe the functional relationship between two quantities from a graph or a verbal description.	Mission 5, Topic B Mission 5, Topic C